MLFF Training-Data and Fine-Tuning Architecture

Purpose and authority

This manual defines the accepted current scientific, statistical, execution, and evidence architecture for the mdstats MLFF workflow: source-certified atomistic data preparation, leakage-safe evidence roles, fitted preparation, multi-view target-subset construction, target-size selection, MACE fine-tuning, protocol validation, deployment, calibration, and bounded campaign execution.

It is intentionally present-tense and single-generation. A reader does not need release chronology, migration history, or obsolete stage semantics to determine current behavior.

The canonical editable architecture sources are the numbered chapters under `docs/arch_manuals/mlff_training_data/`. The assembled Markdown and PDF are generated publication products of those sources and must not be edited as independent authorities.

Detailed exact behavior is owned by current specifications under `docs/specs/training_data/`. Methods/theory material may explain rationale but does not override architecture or specifications. Proposed transitions live in `workplans/`; completed chronology lives under `docs/history/mlff/`; correctness/performance evidence lives in audits, release evidence, and benchmarks.

Architectural motive

MLFF campaigns combine state with fundamentally different epistemic roles: physical source facts, eligibility decisions, evidence partitions, fitted transforms, subset-membership decisions, target-size decisions, optimization/checkpoint state, protocol-validation evidence, calibration evidence, locked tests, and deployment decisions. Conflating those roles creates leakage and ambiguous authority even when the numerical code is correct.

The architecture therefore uses immutable/content-addressed evidence, explicit statistical roles, one normative owner per scientific decision, and authenticated dependency direction. Execution realization is kept separate: cache layout, worker count, queue order, out-of-core storage, and scheduler policy may change without changing scientific membership, ordering, coverage, ranking, or evidence roles.

Expensive exact numerical work is computed once per semantic identity and reused wherever its inputs are unchanged. Exactness, deterministic authoritative decisions, bounded materialization, explicit resource ownership, and restartable authenticated state take precedence over nominal utilization.

Current workflow at a glance

```
source evidence and labels
  -> eligibility / physical conditions
  -> raw feature and event evidence
  -> evidence-role partitioning
  -> fold/final training domains
  -> fitted descriptors, metrics, E0/objective/weight inputs
  -> multi-view feasibility and exact sparse neighborhood authority
  -> MVSEL2 target order
  -> REPAIR2 repaired master order + MVSTATE2 continuation state
  -> independent MVQUAL prefix qualification
  -> configurable target-size study
  -> one protocol-global selected target size
  -> domain-local target prefixes
  -> training / checkpoint selection
  -> held-out protocol validation
  -> final committee / deployment
  -> calibration and activated locked-test / observable validation
```

The current graph has no alternate MVSEL1/REPAIR1 path. Superseded campaign generations are not generally compatible; the sole supported restart boundary is the immediately preceding fixed-fidelity campaign, which is re-authenticated against unchanged upstream preparation inputs before a fresh configurable target-size study is created. Ambiguous or older historical state fails closed at the narrowest required re-preparation boundary.

Reading index

Need	Primary chapter
Scientific motivation, record/evidence model, and scope	Part I - Foundations
Source identity, labels, strain/stress, eligibility, raw features/events	Part II - Data and evidence contracts
Evidence roles, leakage-safe CV, fitted preparation, objective/weighting/exposure boundaries	Part III - Statistical design and fitted preparation
Replay, MACE protocol, checkpointing, validation, deployment, calibration, active learning	Part IV - Training, evaluation, and deployment
FEAS1/MVIDX1/MVSEL2/REPAIR2/MVSTATE2/MVQUAL and target-size study	Part V - Multi-view target subset and size architecture
Exact execution, bounded resource/materialization, cache/restart/storage/progress	Part VI - Performance and execution architecture
Sole-owner matrix and accepted extension boundaries	Part VII - Ownership and extension boundaries
External scientific/algorithmic sources	References

Context retrieval index

For targeted human or AI loading, use the smallest current source containing the needed concept:

Query terms	Load first
source/label identity, eligibility, strain/stress, raw features/events	`20_data_contracts.md`
evidence roles, leakage, CV, fitted metrics, E0, objective, weighting, exposure	`30_statistical_design.md`
replay, MACE, checkpoint, evaluation, deployment, calibration, active learning	`40_training_evaluation.md`
FEAS1, MVIDX1, MVSEL2, REPAIR2, MVSTATE2, MVQUAL, target size, `n1/n2/n3`	`50_target_multiview.md`
scheduler, sparse execution, out-of-core, memory, persistence, progress	`60_execution_performance.md`
owner, dependency direction, unsupported generation, extension boundary	`80_ownership_and_decisions.md`
scientific/algorithmic provenance	`90_references.md`
superseded design rationale or release chronology	`docs/history/mlff/`
proposed transition	`workplans/active/`

Stable terminology

- **training domain** — the DATA5-authorized fold/final gradient-training evidence available to fitted preparation and subset construction.
- **target membership** — frame membership in a target-training subset; owned by MVSEL2/REPAIR2 within a training domain.
- **target size** — the protocol-level scientific target-training cardinality chosen by `TargetSizeStudyPolicy`.
- **monitor size** — the cardinality of a monitoring/evaluation evidence set; never target-size authority.
- **master order** — the one repaired REPAIR2 ordering per training domain whose prefixes define candidate target subsets.
- **qualified size** — a materializable nominal size whose prefix passes independent MVQUAL hard requirements in every required training domain.
- **selected size** — the one protocol-global target size selected from the qualified population.
- **authoritative evidence** — persisted information that defines or independently proves a scientific decision.
- **reconstructible execution cache** — discardable state derivable exactly from authoritative inputs.
- **unsupported generation** — an old campaign/artifact generation that current architecture does not interpret or migrate; it requires re-preparation.

Normative vocabulary

- **SHALL / MUST** — required for scientific, statistical, or execution correctness.
- **SHOULD** — the default design unless measured evidence justifies another exact-equivalent realization.
- **MAY** — optional realization that cannot weaken the scientific contract.

When architecture explains a change-sensitive constant whose exact value is specification-owned, the owning specification remains the sole normative location for changing that value.

Retrieval and local-context rule

Each major chapter states what its concepts own, consume, emit, and explicitly do not own. Equations and symbols are defined near first use. A chapter may repeat a dependency boundary for local comprehension, but repeated prose must not create a second independently tunable contract.

Part I - Foundations

What an MLFF learns

An energy-conserving machine-learned force field represents a potential-energy function

$$E_\theta = E_\theta(\mathbf{Z}, \mathbf{R}, \mathbf{H}),$$

where $\mathbf{Z}$ contains atomic numbers, $\mathbf{R}$ positions, $\mathbf{H}$ the periodic cell, and θ model parameters. Forces and stress follow from derivatives of the same energy,

$$\mathbf{F}_i = -\frac{\partial E_\theta}{\partial \mathbf{R}_i}, \qquad \boldsymbol{\sigma} = -\frac{1}{V}\frac{\partial E_\theta}{\partial \boldsymbol{\epsilon}},$$

under the declared stress sign and strain convention of the label source. MACE constructs symmetry-aware local atomic representations and sums atomic-energy contributions [1]. A useful training/evaluation corpus must therefore constrain both the energy surface and its derivatives throughout the intended simulation domain.

A low global force error is not sufficient. Common framework vibrations can dominate aggregate statistics while rare mobile-ion environments, strain states, migration geometries, interfaces, defects, or other declared focus physics remain poorly represented. The architecture separates broad numerical metrics, condition/group-resolved evidence, physical-observable validation, and explicit extrapolation/challenge evidence.

Why trajectory frames need statistical roles

Molecular-dynamics frames are temporally correlated. Neighboring configurations can be near duplicates, so assigning them to nominally different roles can create leakage and overstate model quality.

For observable x_t, normalized autocorrelation at lag k is

$$\rho_x(k) = \frac{\langle (x_t - \bar{x})(x_{t+k} - \bar{x}) \rangle}{\langle (x_t - \bar{x})^2 \rangle}.$$

A truncated integrated autocorrelation time is

$$\tau_{\mathrm{int},x} = \Delta t \left[\frac{1}{2} + \sum_{k=1}^{k^\star} \rho_x(k) \right],$$

with approximate effective sample count

$$N_{\mathrm{eff},x} \approx \frac{T}{2\tau_{\mathrm{int},x}}.$$

mdstats therefore uses autocorrelation-aware complete-frame blocks, purge semantics, and explicit independence grades rather than treating every frame as independent [3-5]. Exact estimators, truncation, block size, purge, and role-assignment rules are specification-owned.

Evidence-role model

The architecture distinguishes evidence by what it is allowed to control.

Role	Supplies gradients?	May control fitted preparation/subset/size/checkpoint?	Purpose
development / training domain	Yes when selected	Yes, within the authorized training/model-selection contract	fitting and protocol development
checkpoint / common target monitor	No	Yes, only for explicitly authorized development/model-control decisions	stopping/checkpoint and target-size development evidence
held-out CV evaluation	No	No for the frozen protocol it evaluates	protocol validation
calibration	No	No training/subset/checkpoint changes	final-committee uncertainty calibration
locked interpolation/ challenge test	No	No	sealed final evaluation

Calibration is not test data; held-out CV is not a checkpoint monitor; and a monitor cardinality is not a target-training cardinality.

Scope and ownership

The MLFF subsystem owns dataset certification, evidence-role construction, fitted preparation, multi-view target-subset construction, target-size study, training-artifact construction, campaign orchestration, checkpoint/ evaluation lineage, deployment verification coordination, and active-learning lineage.

Its current responsibilities include:

- VASP source discovery/certification and source/label identities;
- composition, thermodynamic condition, ensemble, reference-cell, strain/stress reconstruction;
- electronic-structure compatibility and label-domain grouping;
- energy/force/stress audit and atomic-reference identifiability/fitting lineage;
- immutable frame facts, eligibility, and quality decisions;
- generic raw structural features/events plus explicit optional material/profile extensions;
- autocorrelation-aware complete-frame blocks and role feasibility;
- fixed outer roles and independent CV job families;
- fold/final-domain fitted descriptors, transforms, metrics, E0, objective/weight, and difficulty evidence;
- exact multi-view feasibility/indexing, MVSEL2 ordering, REPAIR2 master order, MVSTATE2 continuation, and MVQUAL qualification;
- one protocol-global target-size decision with domain-local membership;
- MACE target/replay artifacts and explicit exposure realization;
- replay-retention and checkpoint admissibility;
- protocol-matched CV, final training, committee export, calibration, sealed evaluation, and deployment verification;
- active-learning candidate/DFT lineage where supported by current specifications.

The subsystem does not silently merge incompatible electronic-structure levels, infer ambiguous scientific references, use held-out/locked evidence for forbidden model-control decisions, redefine analysis-owned physical-observable algorithms, create a second target selector, generate rescue target sizes, or migrate unsupported old campaign generations.

LTA/zeolite ring, cage, site, crossing, and related semantics are optional profile extensions rather than generic defaults.

Reference application: Li/Na/K-LTA

The principal reference application contains AIMD evidence spanning multiple cation compositions, temperatures, and strain conditions. It motivates—but does not hard-code into generic architecture—several requirements:

1. framework atoms can outnumber mobile cations, so aggregate metrics must not hide declared mobile-species environments;
2. strain conditions need not form a full Cartesian product with composition/temperature, so condition applicability may be hierarchical;
3. one trajectory per condition supplies limited independence and must not be represented as an independent-replica test;
4. fixed framework stoichiometry can make individual atomic reference-energy corrections non-identifiable without anchors;
5. short trajectories may contain few rare transitions, so absent events are explicit coverage gaps rather than evidence of irrelevance.

Reuse of analysis and sampling capabilities

The MLFF workflow orchestrates existing mdstats capabilities instead of duplicating them.

Capability	MLFF use
`mdstats.io.vasp.read_vasp_frames`	cells, coordinates, energies, forces, stress, temperature, provenance
VASP control/ensemble readers	controls, energy-channel and ensemble evidence
trajectory-quality / production-regime assessment	source and stationary-regime evidence
analysis structural/topology modules	optional profile-owned raw evidence or post-training observables under analysis contracts
sampling/cross-fit primitives	source-bound blocks, purge, and independence semantics

Physical observables remain owned by `mdstats.analysis`. The MLFF layer may orchestrate matched evaluation and retain analysis-owned result identities, but it does not redefine RDF, MSD, VACF, VDOS, diffusion, topology, conductivity, or related numerical algorithms.

Current controlling data flow

```
source bytes / controls / trajectory collections
  -> source and label-domain certification
  -> immutable frame facts and eligibility
  -> raw features/events before ordinary thinning
  -> correlation-aware blocks and evidence-role feasibility
  -> development / monitor / CV / calibration / locked roles
  -> required fold/final training domains
  -> domain-local DATA6/DATA7 fitted preparation
  -> FEAS1 + MVIDX1
  -> MVSEL2 -> REPAIR2 / MVSTATE2
  -> independent MVQUAL prefix qualification
  -> common qualified target-size population
  -> target-size study using authorized development/model-selection evidence
  -> one frozen protocol-global N_selected
  -> domain-local selected prefixes
  -> protocol-matched CV with held-out folds inaccessible to size/checkpoint choice
  -> accepted frozen protocol
  -> independent final seeds and checkpoint admission
  -> final committee + deployment artifacts
  -> final-committee calibration where supported
  -> explicit locked-test / observable-validation activation
  -> active-learning lineage where configured
```

No allowed dependency runs from held-out CV or locked-test evidence backward into fitted transforms, E0 fitting, target membership, target-size selection, checkpoint choice, or calibration-policy design.

Responsibility separation is more durable than module layout

The implementation may reorganize Python modules while preserving the architecture. The durable separation is among:

- physical/source facts;
- evidence-role and policy decisions;
- training-domain fitted products;
- target-membership and target-size decisions;
- runtime/execution realization;

- validation/calibration/locked evidence;
- external analysis-owned results.

Current specifications control public/serialized current-generation contracts. Internal refactoring may reuse common sampling/execution primitives when externally owned scientific behavior and persisted current-generation identities remain conforming. Backward compatibility with superseded campaign generations is not an architectural requirement, except for the narrow immediately preceding fixed-fidelity restart boundary: it may reuse authenticated unchanged preparation inputs, but it must create a fresh configurable target-size authority and fails closed when compatibility is ambiguous.

Part II - Data and evidence contracts

Purpose and ownership

This chapter defines immutable source/frame facts, label-domain identity, physical conditions, quality/eligibility, raw feature/event providers, and correlation-aware complete-frame evidence blocks.

It does not own evidence-role assignment beyond the records needed to support DATA5, fitted statistics, target membership, target size, training exposure, checkpoint selection, or validation decisions.

Evidence records and immutability

The MLFF data model separates source facts, workflow decisions, fitted products, runtime realizations, and external scientific results. A new policy creates new policy/decision records rather than mutating immutable source/frame facts.

Source and frame facts

`TrainingDataSource` owns source occurrence identity, path/location hints, content hashes, composition/controls, ensemble/quality/production evidence, label-domain identity, and declared reference grouping.

`TrainingFrameRecord` owns source-bound frame facts such as `frame_uid`, source occurrence, frame index/time, atoms/cell, label references, physical conditions, and distinct geometry/label fingerprints.

`TrainingFrameRecord` does **not** own eligibility, statistical role, target membership, target size, training exposure, calibration, or acquisition state.

Decision, policy, fitted, and realization families

Representative downstream/current families include:

```
FrameEligibilityDecision
PartitionAssignment
CandidateAdmissibilityDecision
AcquisitionDecision

PartitionRoleBudgetPolicy
PartitionFeasibilityReport
FeatureMetricPolicyTemplate
FoldFeatureMetricFit
FinalFeatureMetricFit
TrainingObjectivePolicy
ConfigurationWeightPolicy
PropertyWeightPolicy
CheckpointMetricPolicy
TargetSubsetInputBundle
TargetSizeStudyPolicy
TrainingProtocolIdentity
```

```
MaceCheckpointControlPolicy
ExposureBackendPolicy
MaceExposureRealizationRecord
ReplayRetentionPolicy
ProtocolFreezeRecord
CalibrationApplicabilityDomain
CalibrationTransferDecision
```

A static policy defines an algorithm and fixed choices. A fitted record contains parameters learned from one explicitly authorized training domain. A realization record records behavior actually observed from an external/runtime system. Those roles are not interchangeable.

Target membership is intentionally absent from the generic DATA2-DATA4 record list because its current authority is the MVSEL2/REPAIR2 chain after fitted preparation.

Digests and signatures

Deterministic content/policy/source digests bind identity and detect modification. They do not by themselves authenticate authorship. Serialized current records carry version/schema and deterministic content identity under their owning specifications.

Source manifest and occurrence identity

A review/production manifest supplies source locators, grouping declarations, scientific assertions that cannot be reconstructed unambiguously from one source file, and explicit expert overrides with rationale. Directory/file naming is diagnostic input, not accepted physical truth without verification.

The source byte/content identity is distinct from a manifest occurrence. Byte-identical copies may share a source-content identity while deliberately distinct manifest runs have distinct occurrence identities.

A frame occurrence derives from occurrence identity plus source frame index. This keeps occurrence identity stable across later concatenation/export while permitting duplicate-geometry detection across separate source occurrences.

Geometry, label, and labeled-configuration identities

The architecture keeps three identities separate:

```
geometry_fingerprint
label_payload_digest
labeled_configuration_fingerprint
```

`geometry_fingerprint` identifies atomic geometry independently of energy/force/stress labels under the current canonical wrapping/cell/tolerance policy. `label_payload_digest` binds the selected labeled payload and label-domain identity. `labeled_configuration_fingerprint` combines geometry and label payload.

Leakage auditing may use occurrence overlap, exact geometry overlap, exact labeled-configuration overlap, declared near-geometry/descriptor criteria, restart/copy detection, and forbidden temporal proximity. Approximate or symmetry-aware matching may exist only under an explicit current policy; it cannot change the semantic roles above.

Electronic-structure label domains

Electronic-structure identity is decomposed because not every input difference has the same scientific meaning:

```
TheoryIdentity
EnergyReferenceIdentity
DerivativeConvention
```

```
NumericalQualityProfile
SoftwareProvenance
```

A versioned `LabelCompatibilityPolicy` classifies differences as compatible, compatible with quality evidence, separate label domain, or unresolved. Theory- or reference-defining differences cannot be silently waived.

A target training bundle contains one compatible target label domain and, when enabled, a separately identified replay head/lineage. Incompatible target DFT levels produce separate target bundles rather than an implicit mixed target domain.

Energy channel

The selected target energy is an explicit named channel consistent with derivative labels. Its channel, units, completeness, electronic/reference convention, and provenance are preserved. Energy/force/stress labels that do not share an accepted derivative/reference convention are not silently combined.

Atomic-reference identifiability and fitting boundary

For elemental correction vector $\Delta \mathbf{e}_0$, the schematic fit is

$$\mathbf{A}\,\Delta \mathbf{e}_0 \approx \mathbf{b},$$

with configuration-element count matrix $\mathbf{A}$ and target-minus-foundation energy residual $\mathbf{b}$.

`AtomicReferenceIdentifiabilityReport` depends on elemental count support rather than fitted target residuals. It records element order, matrix shape/rank/singular values, condition/null-space information, identifiable combinations, outcome, and transfer limitations. It does not contain fitted elemental corrections.

The actual `AtomicReferenceFitRecord` is a DATA7 fitted object bound to one fold/final training domain, foundation checkpoint identity, identifiability report, solver/tolerance, elemental support, fitted corrections, residual, and policy outcome.

Each cross-validation fold receives its own fold-local fit. Final training receives a separate final-training fit. Monitors, calibration, held-out evaluation folds, and locked tests are excluded from the fit.

MACE export receives the exact accepted numerical E0 representation, normally an atomic-number mapping; a record/path name is provenance rather than an E0 payload.

Ensemble, temperature, cell, and strain

Ensemble and temperature

The subsystem consumes mdstats control/ensemble certification and distinguishes equilibrium, pressure-controlled, ramped/driven, multi-thermostat, and unresolved cases under the owning control specification. Ensemble is not inferred merely from observed cell variation.

Nominal temperature controls and realized ionic-temperature statistics remain separate. `TemperatureCondition` binds requested/thermostat targets, realized series/statistics, drift/stationarity evidence, and ramp status as applicable.

Reference-cell resolution

Strain requires an explicit or uniquely resolvable compatible reference. Accepted resolution order is controlled by current specifications and may use an explicit matrix/structure/run or a unique compatible unstrained member of the declared reference group. Ambiguity fails closed.

Cell and deformation convention

ASE cell vectors are rows. For fractional row vector $\mathbf{s}_{\mathrm{row}}$ and cell $\mathbf{H}$,

$$\mathbf{r}_{\text{row}} = \mathbf{s}_{\text{row}}\mathbf{H}.$$

For reference $\mathbf{H}_0$ and current cell $\mathbf{H}_t$, the reported deformation gradient acting on Cartesian column vectors is

$$\mathbf{F}_t = \left(\mathbf{H}_0^{-1}\mathbf{H}_t\right)^T.$$

An internal right-acting row-vector form is acceptable only when serialization/reporting returns the declared Cartesian-column convention. Rotation/stretch separation uses the declared polar-decomposition convention.

Stored strain evidence includes the applicable volume, linear/finite/logarithmic, hydrostatic/deviatoric, principal, shear, rotation, coordinate-frame, and storage-convention quantities. Qualification includes nonsymmetric shear and rotated-stretch cases so transpose/left-right errors cannot hide behind diagonal fixtures.

Hierarchical condition schemas

Condition space is not assumed to be a global Cartesian product. A material profile declares applicable condition axes and hierarchical strata. For example, an LTA profile may separate unstrained composition/temperature/regime strata from strained composition/reference-condition/strain-mode/sign/regime strata. Only observed and scientifically applicable combinations are required.

Stress and virial

Canonical `REF_stress` is a symmetric Cartesian Cauchy-stress tensor in eV/Angstrom^3 under the ASE/MACE sign convention qualified by the runtime lock. Tensor shear carries no engineering-factor multiplication. Intermediate Voigt ordering, when used, is explicit and round-tripped.

Virial and stress have distinct keys and are never silently relabeled. Qualification covers units, finite-strain sign, tensor/Voigt order, shear factors, and MACE read-back. Missing stress may carry zero stress weight only under an explicit heterogeneous-label policy.

Eligibility and quality

Run/source quality distinguishes qualified, degraded, unqualified, and unresolved states under current policy. Overrides are explicit evidence.

`FrameEligibilityDecision` applies after labels exist. Hard rejection includes absent/nonfinite required labels or geometry/cell, singular/corrupt structures, incomplete ionic records not recoverable under the current interruption policy, catastrophic overlaps, and disallowed electronic-convergence failures.

Soft evidence records transient regimes, unusual but physical forces/stress, rare coordination/events, topology changes, model residuals, and degraded numerical quality without turning percentile tails into automatic rejection.

Pre-DFT candidates use a separate `CandidateAdmissibilityDecision` over geometry/cell safety, element/count policy, topology/integrity, trajectory/integrator evidence, model outputs, and descriptor availability. After DFT labeling they re-enter normal source/frame eligibility lineage.

Material profiles and feature providers

Declarative profile boundary

`SystemProfileProvider` owns material identity: phases, geometry, chemistry modifiers, optional structural extensions, meaningful atom groups, condition axes, and independence axes. It does not itself own calculated scientific feature arrays.

Profiles are compositional rather than a single flat material enum. Interface/multiphase systems explicitly declare component membership. Generic fallback supplies only generic groups/axes; porous/zeolite/LTA semantics require the corresponding explicit extension chain and never activate automatically.

Universal structural selection inputs

The generic structural provider supplies selection-grade local geometry descriptors such as smooth chemistry-scaled coordination, support-neighbor count, radial projections, local-density/mixing proxies, angular moments, and rotationally invariant orientational-order summaries.

These are upstream target-subset inputs, not an independent selector and not replacements for analysis-owned RDF, integer coordination, full angle distributions, structure factors, or topology observables.

Descriptors aggregate by authorized atom groups and elements present in the permitted domain. Generic temporal events capture large local structural changes without assigning material-specific physical meaning.

Partition-critical profile features

Rare categorical states that partition policy promises to protect are available at full resolution before the outer partition freezes. A profile may supply phase, environment, defect, region, molecular, or event states.

Optional LTA state includes framework/mobile roles, resolvable ring/site class, off-center class, coordination/site/ring-crossing changes, and framework-integrity evidence. Unresolved required classifications produce explicit coverage/partition limitations rather than fabricated balanced strata.

Optional learned-model features

A qualified optional MACE provider may supply foundation/model identity, invariant atomic descriptors, group/species environment summaries, and authorized zero-shot prediction/residual evidence. MACE/PyTorch remain optional dependencies to the mdstats core.

Feature blinding and fitted metrics

Geometry-only descriptors from a frozen model may be computed wherever authorized. Label-derived residual/difficulty features may be exposed only inside their applicable training domain.

Outer monitor, calibration, held-out evaluation, and locked-test residuals do not enter feature fitting or target-subset construction. Evaluation predictions can be persisted in blinded catalogs without exposing residual-derived selector inputs. Violating this boundary is a hard leakage failure.

Raw feature providers are partition-independent. Dataset-dependent scaling/PCA/whitening/metric fitting is represented by separate static templates and fold/final fitted records. For fold k, fitting may inspect only its gradient-training domain; other domains may be transformed by the frozen fitted object but cannot influence it.

A block-normalized metric may take the form

$$d^2(i,j) = \sum_b w_b \frac{\left\| \mathbf{z}_i^{(b)} - \mathbf{z}_j^{(b)} \right\|_2^2}{d_b},$$

where block weights, dimensions, missing-block behavior, dtype, tolerance, and any fitted scaling/projection are explicitly identified. High-dimensional learned descriptors do not dominate solely because they contain more components.

Event detection before thinning

The controlling order is:

1. source/label integrity on full-resolution frames;

2. event/change detection on all eligible frames;
3. protected event-window preservation;
4. temporal thinning of the ordinary non-event pool;
5. higher-cost descriptor/fitted target-subset-input operations.

Event stencils are policy-controlled and compact by default. Adjacent frames from one physical event are not mistaken for independent rare-event evidence.

Autocorrelation and complete-frame blocks

Fast observables determine the minimum ordinary decorrelation/block scale; slow structural variables diagnose whether state-level independence exists at all. Candidate stride is defined in physical/frame time from the declared fast autocorrelation estimate and applies only to the non-event pool.

A `TrainingDataBlock` is a contiguous interval of whole configurations with block/run/frame bounds, represented time, regime, correlation evidence, and configuration identities. Atoms from one configuration are never split across statistical roles.

A purge interval separates roles using the declared physical/autocorrelation/event/restart policy. If a slow state never decorrelates, the independence report explicitly states that temporal blocking does not provide state-level independence.

Part III - Statistical design and fitted preparation

Purpose and ownership

This chapter defines the statistical evidence roles that make later model comparisons interpretable and the fitted-preparation boundary immediately upstream of multi-view target-subset construction.

It owns the architectural separation among training, monitoring, cross-validation, calibration, and locked-test evidence. It also defines what fold/final-domain fitted preparation may consume and emit.

It does **not** own target membership or target size. DATA7 prepares fitted inputs; MVSEL2/REPAIR2 determine target membership inside each authorized training domain; `TargetSizeStudyPolicy` chooses the protocol-global target size.

Independence and evidence roles

Independence hierarchy

Evidence uses the strongest available independence level, for example:

1. independent replica/velocity seed or independently prepared realization;
2. independent structural/chemical ordering;
3. independent thermodynamic run;
4. purged temporal block within one run.

Temporal separation does not create an independent metastable state when the relevant slow variable never decorrelates. Every cohort carries machine-readable independence evidence and known limitations.

Partition feasibility

Before assigning roles, the partition policy declares requested cohorts, cross-validation support, minimum independent blocks/grades, purge requirements, and allowed reductions. A feasibility report states what the available evidence can support.

Valid outcomes include full support, temporal-block-only support, deferred calibration, external-only challenge evidence, reduced fold count, or insufficient support. The workflow never fabricates every desired role from a short or correlated trajectory to satisfy a percentage target.

Outer evidence roles

A feasible target label domain may contain:

```
development_pool
outer_monitor_validation
uncertainty_calibration
locked_interpolation_test
zero or more locked_challenge_tests
```

Only the development pool supplies gradient-training candidates. The common target monitor is development/model-selection evidence: it may control the current authorized monitoring/checkpoint and target-size-screen policies but supplies no gradients and is not a held-out CV fold or locked test.

Calibration is reserved for predictions from the actual final committee. Locked interpolation/challenge evidence cannot affect fitting, subset construction, target-size selection, training protocol, checkpoint selection, calibration-policy choice, acquisition policy, or protocol design.

When a requested role is unsupported, it is absent/deferred explicitly rather than synthesized from correlated evidence.

Cross-validation validates a frozen protocol

A frame that supplied a gradient is not independent validation evidence for that model. Likewise, a held-out evaluation fold cannot control stopping, checkpoint choice, or target-size choice for the protocol it is intended to evaluate.

For cross-validation fold k, keep distinct:

```
fold_training_domain_k
fold_checkpoint_monitor_k
held_out_evaluation_fold_k
```

The fold model has fresh model/optimizer/checkpoint lineage. Fitted transforms, feature metrics, E0 fits, difficulty evidence, and target membership are constructed only from `fold_training_domain_k`. Checkpoint selection uses its authorized monitor, not the held-out evaluation fold.

Target size is frozen **before** protocol-matched held-out CV evaluation. The same selected cardinality is used as a protocol hyperparameter across required folds/final development, while each domain has its own leakage-safe target membership. Only after checkpoint choice freezes is a fold model evaluated on `held_out_evaluation_fold_k`.

Cross-validation is therefore a family of independent jobs evaluating one frozen protocol, not a rotating epoch schedule and not an inner loop for choosing target size.

Fitted preparation inside each training domain

For each fold/final training domain, DATA6/DATA7 may construct products whose statistical meaning depends on that domain. These include, as applicable:

- descriptor transforms and heterogeneous fitted feature metrics;
- training-domain foundation-model predictions/residual difficulty evidence;
- atomic-reference/E0 fits;
- objective, configuration-weight, and property-weight records;
- condition/provenance/event/environment/diversity inputs needed by target-subset construction;

- deterministic identities binding those products to the training domain and complete protocol.

No fitted product may inspect held-out CV evaluation, calibration, or locked-test evidence unless an owning specification explicitly gives it a non-training role that preserves the relevant independence boundary.

Raw versus fitted information

Partition-independent physical facts and raw feature/event providers belong upstream. A fitted normalization, metric, model residual, E0 correction, or difficulty transform belongs to the training domain that fitted it.

This distinction prevents an apparently innocuous global normalization or residual calculation from leaking held-out evidence into subset construction.

Selection inputs are not a second selector

Representative density, diversity/FPS, environment coverage, condition balance, protected events, difficulty, provenance/correlation structure, and mandatory anchors remain useful scientific information. They do not define an independent DATA7 target order.

DATA7 expresses them as one or more of:

```
fitted feature coordinates/metrics
hard obligations or applicability masks
representative-density / utility evidence
diversity evidence
event/environment/condition evidence
difficulty evidence
correlation/provenance identities
policy inputs with deterministic identities
```

The one current membership authority consumes these inputs in the multi-view chain described in Part V. There is no competing quota/FPS `TrainingSelectionPlan` whose prefixes can disagree with MVSEL2/REPAIR2.

Material/profile specialization

Condition axes and focus groups are declared by the applicable material/profile contract. They may include composition, temperature, pressure, strain, phase, defect, surface/interface state, molecular conformer, preparation history, or other scientifically justified axes.

A profile may define hierarchical applicability rather than a global Cartesian product. Empty or physically inapplicable combinations are not treated as missing observations merely because all axis names exist.

Material-specific semantics remain explicit extensions. Li/Na/K focus groups, ring/cage/site concepts, or LTA-specific condition hierarchies are not generic defaults.

Training objective, weighting, and exposure

Target membership, target size, loss weighting, and runtime exposure are separate decisions.

`TrainingObjectivePolicy` binds the loss family, energy/force/stress weights, head weights, normalization, robust-loss choices, and missing-label behavior. Configuration/property weighting policies bind condition/regime/event/quality and property-specific weights.

Exposure realization binds the head, eligible use, actual gradient exposures, configuration/property weights, sampling/duplication behavior, seed, and runtime lineage as applicable.

A frame can therefore be selected once, weighted non-uniformly, and exposed according to a qualified loader policy without those three decisions becoming the same authority.

Atom-group force imbalance

A configuration can contain many more host force components than scientifically critical minority-group components. Subset diversity alone does not eliminate that loss imbalance.

The standard MACE configuration/property-weight path does not claim a generic atomwise group-weighted loss. Evaluation/checkpoint policy therefore reports declared group-resolved metrics and imposes applicable group constraints. A custom atomwise or auxiliary loss defines a different `TrainingProtocolIdentity` and requires separate qualification.

Exposure backends

The qualified fixed-file MACE path materializes selected target/replay frames in fixed artifacts and binds the realized upstream loader shuffle/batching behavior into the protocol.

Dynamic epoch resampling, multi-job resampling, or alternate final-refit exposure semantics are valid only when a current adapter/specification supports them and records optimizer/checkpoint/exposure lineage. Static files alone cannot claim dynamic resampling.

Realized exposure audit

Exposure evidence compares intended artifacts and weights with observed loader behavior, including target/replay counts, implicit duplication, expected/observed batches, and configuration/property exposures.

Upstream target/replay duplication behavior is version-dependent. The adapter either disables unintended duplication when supported or binds the realized behavior into the protocol and verifies it. Silent changes in effective target/replay exposure fail closed.

Statistical dependency boundary

The allowed dependency direction is:

```
raw source / feature / event evidence
    -> role assignment
    -> training domain
    -> fitted preparation
    -> MVSEL2 / REPAIR2 target membership
    -> target-size study using authorized development/model-selection evidence
    -> frozen target size and training protocol
    -> checkpoint selection
    -> held-out protocol validation
    -> calibration / locked-test activation
```

Forbidden reverse dependencies include:

- held-out CV error choosing target size;
- locked-test evidence tuning subset policy or checkpoint policy;
- calibration evidence fitting the training protocol it calibrates;
- execution worker/cache/scheduler behavior changing partition membership, fitted domains, target order, or evidence roles.

This dependency boundary is the statistical contract that makes the later model-quality evidence interpretable.

Part IV - Training, evaluation, and deployment

Purpose and ownership

This chapter defines the current training-protocol identity, replay boundary, checkpoint admissibility, protocol-matched cross-validation, final training/committee construction, calibration, sealed evaluation, deployment verification, and active-learning lineage.

Target membership and target size are already frozen by the Part V authorities before protocol-validation cross-validation is interpreted. This chapter consumes those decisions; it does not create a second subset or size authority.

Multi-head replay and complete protocol identity

Multi-head replay fine-tuning trains a shared MACE backbone on target data and a foundation replay dataset with separate output heads. Replay constrains catastrophic forgetting while the target head adapts.

Every scientifically compared run is bound to a complete `TrainingProtocolIdentity` containing, as applicable:

```
foundation checkpoint / model family / head identity
selected protocol-global target size
domain-local target-membership identity
replay source, split, and replay-monitor identity
training objective and property/configuration weights
target/replay head weights
exposure backend and realized balancing/duplication policy
checkpoint metric and checkpoint-control policy
replay-retention policy
optimizer, LR schedule, epoch cap, stopping policy, and seed policy
model precision and execution backend
MACE adapter/runtime lock
```

Cross-validation evidence validates only the protocol identity it actually used. A change to replay semantics, objective, selected size, membership policy, checkpoint policy, precision/backend, stopping/LR policy, or another protocol-defining field creates a different protocol.

Separate target and replay evidence

Target and replay evidence retain separate source/label identities, atomic-reference rules where applicable, split/membership plans, weights/exposure accounting, and monitors. Replay training and replay monitoring are disjoint evidence roles.

The mdstats workflow records replay preparation and does not silently acquire external replay data. True-label replay is evaluated against held-out labels; pseudo-label replay, when supported, measures drift from the bound foundation model on an unseen sentinel set.

`ReplayRetentionPolicy` binds the retention metric, baseline, allowed degradation, aggregation, and failure semantics. A checkpoint that violates a mandatory replay-retention requirement is inadmissible even when its target metric improves.

Common online monitors

Monitoring evidence sets are deterministic protocol inputs with their own policy identities. Their cardinalities are monitor properties, not target-size candidates.

The common target monitor is authorized development/model-selection evidence. It may be used by the target-size study and by the current checkpoint/stopping policy as explicitly specified. It supplies no gradients and is distinct from held-out CV evaluation and locked tests.

The replay monitor is separately owned and separately identified. Numeric equality between a monitor cardinality and one nominal target size has no semantic effect.

Checkpoint metrics and constrained choice

`CheckpointMetricPolicy` defines the primary target objective and every mandatory target, focus-group/species, condition, energy/stress/property, replay, and physical-integrity constraint applicable to checkpoint admission.

A typical constrained form is

$$\min_{c} L_{\text{target monitor}}(c)$$

subject to requirements such as

$$L_{F,g}(c) \leq \delta_g, \qquad \Delta L_{\text{replay}}(c) \leq \delta_{\text{replay}}.$$

Exact metrics and thresholds are specification-owned serialized policy. Replay retention, structural integrity, relaxation/deployment integrity, and similar mandatory predicates are constraints rather than score bonuses unless an explicit current policy says otherwise.

Checkpoint selection is deterministic over the complete authorized candidate set and fails closed when no candidate satisfies mandatory constraints.

MACE adapter and runtime lock

The current MACE adapter binds the upstream behaviors on which the protocol depends, including package/source identity, target/replay head ordering, loader realization, scheduler/stopping behavior, checkpoint retention, precision/backend realization, and accelerator qualification where applicable.

Documentation URLs are not a runtime contract. If version-locked upstream behavior changes materially, preparation or qualification fails closed until the current adapter specification is revised and requalified.

Minimal Extended XYZ plus sidecar provenance

Extended XYZ contains only MACE-readable labels, weights, and compact stable identities. Long provenance, policy identities, and audit reasons live in sidecar manifests keyed by stable frame/configuration identity.

Target export includes the declared energy channel, forces, authorized stress, configuration/property weights, cell/PBC, atom order, and exact label-domain/E0 provenance. Export precision and round-trip behavior are qualified through the current parser/reader path.

Explicit E0 realization

An `AtomicReferenceFitRecord` is converted to the exact numerical representation accepted by the current MACE runtime, normally an explicit atomic-number mapping. A provenance record name or path never substitutes for the numerical E0 payload.

Label-domain boundary

A target bundle contains one compatible target `LabelDomain` and, when replay is enabled, a separately identified replay head/lineage. Incompatible target electronic-structure domains are not silently merged.

Target-size study versus ordinary stopping

The target-size experiment is a special protocol-comparison control described in Part V. It uses authenticated `n1 -> n2 -> n3` continuation at exact configured boundaries, with a common seed set, and disables ordinary target-success early stopping so candidate sizes reach comparable fidelity boundaries. Where TRAIN2 needs a full deterministic schedule extent, it derives that value from the terminal boundary; it is not a second target-size authority. The separate production maximum `n` is reserved for a fresh selected-size campaign. Hard numerical/scientific failure remains a valid rejection.

Epoch has deliberately different semantics in the two phases. During target-size selection, epoch is a **controlled variable**: the configured coarse, short, and final screens consume only exact `n1`, `n2`, and `n3` checkpoints. An earlier checkpoint is inadmissible even when it scores better, because substituting it would confound target-data size with achieved training fidelity. The public `select-target-size` operation owns this complete restartable `n1 -> n2 -> n3` experiment; generated campaigns default to `(n1,n2,n3)/n = (1,3,10)/30`, with `n` consumed only by fresh post-selection production.

After `N_selected` is frozen, ordinary production/CV training resumes under the frozen protocol. Production checkpoint epoch is then a **selectable model variable**: production `evaluate` may choose an earlier admissible checkpoint when it is better under the frozen checkpoint-selection policy, even though the configured training horizon remains `n` epochs. Its target-oriented stopping and LR-refinement semantics are part of `TrainingProtocolIdentity`; changing them after protocol comparison invalidates the comparison.

The stable TRAIN2 command boundary is therefore `prepare -> preflight -> select-target-size -> materialize -> preflight -> train -> evaluate -> verify`. `prepare` owns only the initial screening workload; `materialize` owns only the selected-size final-development/CV realization; both `preflight` occurrences have the same operational meaning and are bound to the exact current DATA8 matrix. The screening preflight remains valid throughout an unchanged `n1/n2/n3` candidate matrix, while selected-production materialization changes that matrix and therefore requires a new preflight.

Gate TRAIN2B

TRAIN2B executes one authenticated trajectory per `(target size, seed)`. During screening it durably pauses only at the active exact boundary, then the real target-size owner ranks outcomes before authorizing survivors to continue. Continuation preserves model parameters, EMA state, optimizer/LR state, and Python/NumPy/Torch CPU/CUDA RNG states. `train2_true_replay` remains a bounded runtime monitor below this scheduler/selection boundary. Restart restores live non-EMA parameters, EMA state, optimizer/LR state, and RNG ancestry before new work. A run that has passed its active boundary is invalidated to a fresh coarse screen; it cannot supply current ranking evidence. Eliminated-size jobs receive no later authorization.

Protocol-matched cross-validation

Cross-validation validates the **complete already-frozen protocol**, including selected target size. It does not choose target size.

For each fold k:

1. DATA5 provides `fold_training_domain_k`, a disjoint authorized checkpoint monitor, and `held_out_evaluation_fold_k`.
2. DATA6/DATA7 fit descriptors, transforms, metrics, E0, objective/weights, and difficulty evidence only within `fold_training_domain_k`.
3. MVSEL2/REPAIR2 construct the fold-local repaired master order from fold-authorized evidence.
4. The already-frozen protocol-global `N_selected` defines the fold target prefix.
5. A fresh model/optimizer lineage is trained under the bound production stopping/checkpoint policy.
6. Checkpoint choice freezes without inspecting `held_out_evaluation_fold_k`.
7. Only then is the checkpoint evaluated on the held-out fold.

The fold membership is local because each fold has different authorized evidence; the selected cardinality is global because it is part of the one protocol being validated.

If held-out fold performance were used to select `N_selected`, that evidence would no longer be independent protocol validation unless the complete size-selection procedure were nested inside another outer validation design.

Final training and committee construction

After protocol-matched CV is accepted, final-development fitted products and the final-domain target master order are already governed by the same frozen protocol and selected size. Final seeds are trained independently under that protocol.

Candidate checkpoints are evaluated under the current constrained policy. The selected target heads are exported and a committee is constructed with explicit member/seed/checkpoint identity.

`ProtocolFreezeRecord` binds the final training protocol, selected target-size decision, final-domain target-membership identity, replay/monitor identities, model/checkpoint identities, committee identity, and required upstream evidence.

Sealed evaluation and deployment

Development artifacts are separated from calibration and sealed-evaluation artifacts. A locked evaluation bundle may exist before activation, but development/training/checkpoint processes cannot inspect it.

Locked-test activation requires the frozen protocol/committee plus every owning-specification promotion predicate. Locked evidence cannot retroactively alter fitted preparation, target membership, target size, stopping/LR policy, checkpoint selection, replay policy, calibration-policy choice, or acquisition policy.

Deployment artifacts are produced only from admitted final target heads with explicit precision/runtime identity. Deployment verification is bounded and uses the frozen downstream-runtime contract. Structural/relaxation failure, NaN/Inf behavior, topology breakage where prohibited, or another mandatory deployment-integrity failure rejects the candidate independently of force-RMSE rank.

Calibration and uncertainty lineage

Committee disagreement is a ranking signal, not an error guarantee. Numerical uncertainty/acquisition thresholds are calibrated only using predictions of the actual frozen final committee on an authorized calibration cohort.

Calibration identity binds model/committee digests, complete training protocol, target/replay/seed/runtime lineage, precision/backend, calibration cohort, and declared applicability domain.

A transfer decision distinguishes at least:

```
within_calibrated_domain
rank_only_outside_domain
recalibration_required
rejected_incompatible_domain
```

Without valid final-committee calibration, acquisition is explicitly uncalibrated or rank-only. Locked tests are excluded from calibration and acquisition.

Active-learning lineage

Selection-biased active-learning labels enter a new development/training candidate pool. Existing frame roles are inherited unchanged by default. Independent new evidence may create new calibration/validation/challenge cohorts only through explicit lineage.

Repartitioning previously classified evidence creates a new evaluation lineage rather than silently rewriting old roles. A new active-learning generation may require re-preparation of fitted products and target membership; this is normal current-generation construction, not compatibility migration of obsolete campaign schemas.

Reproducibility identity

A reproducible campaign binds, as applicable:

- source/parser and label-domain identities;
- partition/independence roles;
- feature/provider and fitted DATA6/DATA7 product identities;
- MVIDX, MVSEL2, REPAIR2, MVSTATE2, and MVQUAL identities;
- target-size decision and domain-local target-prefix identities;
- foundation/model/runtime lock;
- replay and monitor identities;
- objective, weights, exposure realization;
- optimizer/LR/stopping/seed policy;
- checkpoint metrics/admission decision;
- committee/protocol freeze;
- calibration and locked-test activation evidence;
- output/deployment checksums.

Execution-only worker counts, queue completion order, cache paths, file-backing choice, and similar non-semantic settings are excluded unless a current specification explicitly declares otherwise.

Failure semantics

The workflow fails closed when, among other current-specification conditions:

- source/label identity is unresolved or incompatible;
- required strain/reference conventions are ambiguous;
- requested evidence roles are infeasible under the declared independence policy;
- held-out, calibration, or locked evidence reaches a forbidden fitted/subset/size/checkpoint operation;
- a fold held-out evaluation controls checkpoint choice or target size;
- compared CV and final runs do not share the claimed complete protocol identity;
- runtime behavior differs materially from its qualified lock;
- realized target/replay exposure differs from accepted protocol;
- no checkpoint satisfies mandatory target/focus/replay/integrity constraints;
- calibrated acquisition is attempted outside its applicability domain without the declared transfer action;
- active-learning lineage silently rewrites prior evidence roles.

Absent rare events, replicas, condition combinations, calibration cohorts, or challenge sets are reported as limitations/coverage gaps rather than fabricated evidence.

Part V - Multi-view target subset and target-size architecture

Purpose and ownership

This chapter defines how an authorized fold/final training domain becomes one deterministic nested family of target-training subsets and how one protocol-global target size is selected without consuming held-out validation evidence.

The current chain is:

```
DATA7 fitted selection inputs
    -> FEAS1 full-pool feasibility
    -> MVIDX1 exact sparse neighborhood authority
    -> MVSEL2 progressive target order
    -> REPAIR2 repaired master order / MVSTATE2 continuation state
```

```
-> MVQUAL independent prefix qualification
-> TargetSizeStudyPolicy
-> selected target size
```

Each component has one role. MVSEL2 is the only current ordering authority; REPAIR2 is the only current repair authority; MVSTATE2 is the only current continuation-state family; MVQUAL independently verifies hard requirements. Superseded selector/repair/migration generations are historical and are not alternate current paths.

Why multi-view subset construction is necessary

A target subset must cover several physically meaningful feature views simultaneously. Optimizing one descriptor distance or one average utility can conceal a severe deficit in another required view. The architecture therefore treats each required family as an explicit coverage relation, diagnoses full-pool feasibility before subset optimization, and maintains hard coverage separately from discretionary representative utility.

Four principles control the design:

1. feasibility precedes subset optimization;
2. hard coverage and obligations cannot be traded away by aggregate score;
3. subset ordering is deterministic and nested so size comparisons are not confounded by resampling;
4. selector/repair state and independent qualification evidence are separate authorities.

Exact multi-view neighborhood authority

For feature family m, let $x_w^{(m)}$ be witness coordinates, $x_c^{(m)}$ candidate coordinates, D_m the frozen scaling transform, and $r_w^{(m)}$ the authoritative witness radius. Exact adjacency is

$$A_{wc}^{(m)} = \mathbf{1}\left[\left\| D_m\left(x_w^{(m)} - x_c^{(m)}\right) \right\|_2 \le r_w^{(m)}\right].$$

For selected subset S, witness multiplicity is

$$n_w^{(m)}(S) = \sum_{c \in S} A_{wc}^{(m)},$$

and weighted family coverage is

$$C_m(S) = \frac{\sum_w \omega_w^{(m)} \, \mathbf{1}[n_w^{(m)}(S) > 0]}{\sum_w \omega_w^{(m)}}.$$

For hard threshold τ_m owned by the current coverage policy, family deficit is

$$D_m(S) = \max(0, \tau_m - C_m(S)),$$

and the worst-view deficit is

$$D_{\max}(S) = \max_m D_m(S).$$

A weighted average across families cannot substitute for a failed required family.

Approximate-neighbor substitutions are not scientifically equivalent to the exact relation unless a future accepted design explicitly changes that contract.

Full-pool feasibility (FEAS1)

FEAS1 inspects the complete eligible candidate/reference authority before subset ordering begins. It answers whether the frozen hard coverage and obligation predicates are satisfiable at all and how fragile that support is.

For witness w,

$$d_w^{(m)} = \sum_{c \in \mathcal{C}} A_{wc}^{(m)}.$$

Low-support witness mass and mandatory-obligation incidence identify regions where correlation-unit exclusion, provenance restrictions, or subset-size ceilings can destroy feasibility.

FEAS1 may conclude that a requested rung cannot satisfy the frozen predicates. That conclusion is evidence about the data/policy combination, not permission to relax coverage, fabricate candidates, or invent an intermediate target size.

One exact sparse relation (MVIDX1)

MVIDX1 owns the authenticated exact sparse neighborhood relation used by selection, repair, and qualification. Scientific identity binds at least:

- candidate and witness identity/order;
- feature-family identity;
- scaling/distance/radius semantics;
- exact sparse cardinalities/content;
- policy-relevant schema identity.

Execution-only details such as worker count, query block size, in-memory versus file-backed inversion, mmap placement, queue depth, or NUMA placement do not change scientific identity.

The selector/repair path consumes a forward-oriented projection sufficient to score candidate rows and obligation/correlation incidence. Independent qualification may consume the authenticated primitive relation it needs, but no downstream component recomputes geometry under a competing numerical implementation when the same semantic MVIDX authority already exists.

Large exact inversions may use deterministic out-of-core execution as described in Part VI.

Progressive target ordering (MVSEL2)

MVSEL2 constructs one deterministic progressive order π_d for training domain d. It consumes DATA7 fitted selection inputs, FEAS1 evidence, MVIDX1, hard obligations, correlation/provenance structure, and the frozen selector policy.

Scientific priority structure

The exact lexicographic policy is specification-owned. Architecturally it has two classes of responsibility:

1. satisfy hard/worst-view coverage and mandatory obligations first; and
2. once hard requirements permit, improve representative utility/diversity and declared balance objectives without violating the hard state.

Representative density, diversity/FPS, environment coverage, protected events, condition balance, difficulty, and provenance are inputs to this one selector rather than independent target-membership authorities.

Deterministic exact execution

Rank authority is sequential because the scientific state changes after every accepted candidate. Parallel or vector execution may accelerate preparation and exact candidate scoring, but the authoritative accepted candidate and FP64 scientific decision semantics must remain unchanged.

The current execution architecture uses compact witness/obligation/correlation state and exact on-demand forward-row scoring rather than a product-scale eager inverse marginal array for every candidate. During hard-coverage selection, exact staged filtering may eliminate candidates only when the frozen lexicographic ordering proves they cannot win.

After hard coverage completes, an exact certified lazy representative frontier may avoid full rescoring. Stale upper bounds are conservative; candidates are refreshed until the winner is certified under the frozen tolerance and complete tie hierarchy. A full-forward exact oracle remains a bounded correctness fallback, not a separate scientific policy.

Exact repaired master order (REPAIR2)

MVSEL2 produces the progressive order; REPAIR2 is the sole authority allowed to perform the declared exact active-shell repair while preserving protected lower prefixes and hard invariants.

For selected candidate c, unique covered mass is

$$U(c \mid S) = \sum_{m} \sum_{w:A_{wc}^{(m)}=1} \omega_w^{(m)} \, \mathbf{1}[n_w^{(m)}(S) = 1].$$

A removal is admissible only when it satisfies the frozen unique-support and hard-obligation safety predicates. Replacement proposals follow the frozen deficit/frontier objective and deterministic tie hierarchy. Accepted swaps cannot regress protected hard coverage.

Proposal scoring within one immutable pre-swap state may execute concurrently; authoritative winner comparison and state mutation remain deterministic.

REPAIR2 publishes **one repaired master order per domain**. Candidate target subsets are prefix views of that order. The architecture forbids independently repairing separate copies of the 128-, 256-, 512-, or other rungs because that would destroy nesting and make size comparisons conflate cardinality with unrelated membership changes.

Reconstructible continuation state (MVSTATE2)

MVSTATE2 is compact authenticated continuation state for the current selector/repair generation. It binds the dataset/domain identity, UID/family order, MVIDX identity, weights, obligations, correlation units, selector/repair policy, selected prefix, and current schema identity.

It persists only state required to reconstruct the current scientific position, such as witness multiplicity, covered mass, obligation/correlation counts, and representative utility. Product-scale complete candidate marginal arrays and ephemeral lazy-heap contents are not authoritative continuation state.

Publication is atomic. Restoration revalidates the persisted state against the selected prefix and primitive identities. Incompatible historical state is not migrated into MVSTATE2; the current campaign must reconstruct from current authoritative inputs or be re-prepared.

Independent prefix qualification (MVQUAL)

MVQUAL independently recomputes the hard coverage and obligation evidence required to qualify candidate prefixes. It records policy-defined evidence such as:

- per-family hard deficits;
- uncovered weighted mass and counts;
- redundancy/unique-support diagnostics;
- mandatory-obligation satisfaction;
- provenance/correlation diversity diagnostics;
- exact input and policy identities.

Selector/repair internal counters are not accepted as independent qualification evidence. MVQUAL may share authenticated primitive sparse inputs, but it recomputes the relevant predicates through its independent verification path.

Locked-test data cannot tune radii, weights, hard thresholds, repair budgets, tie rules, or qualification predicates.

Nested prefixes and hard-coverage monotonicity

Let π_d denote the repaired master order for domain d. The target subset at size N is

$$D_{d,N} = \pi_d[:N].$$

For $N_2 > N_1$,

$$D_{d,N_1} \subset D_{d,N_2}.$$

Under fixed exact hard-coverage and obligation predicates, adding candidates cannot remove already-covered witnesses or already-satisfied positive obligations. Therefore hard satisfaction cannot regress solely because N increases.

A pass/fail/pass qualification pattern across increasing prefixes is an invariant violation. It indicates broken nesting, identity, qualification logic, obligation semantics, or numerical realization and must fail closed.

Target-size populations

The scientific target-size study uses a fixed nominal population

$$\mathcal{N}_0 = \{128, 256, 512, 1024, 2048, 4096, 8192, 16384\}.$$

For required training domain d,

$$N_{\text{available},d} = |\mathcal{D}_{\text{eligible},d}|.$$

The common materializable population is

$$\mathcal{N}_M = \left\{N \in \mathcal{N}_0 : N \le \min_d N_{\text{available},d}\right\},$$

where required domains include final development and every required cross-validation gradient-training domain.

Independent MVQUAL evidence defines

$$\mathcal{Q} = \{N \in \mathcal{N}_M : \text{all hard requirements pass in every required domain}\}.$$

The selected size satisfies

$$N_{\text{selected}} \in \mathcal{Q} \subseteq \mathcal{N}_0.$$

No dynamic rescue size is created. An arbitrary available-pool cardinality, monitor cardinality, replay cardinality, batch size, or implementation budget can never silently become a scientific target size.

Domain-local membership, protocol-global size

Each required training domain has its own leakage-safe fitted inputs and repaired master order. Therefore the actual selected frames differ by domain even when the selected cardinality is common.

`N_selected` is one protocol hyperparameter shared across required fold/final jobs. This permits protocol-matched cross-validation without leaking held-out fold evidence into size selection.

Held-out cross-validation folds evaluate the complete already-frozen protocol. If held-out fold performance were used to choose `N_selected`, those folds would no longer be independent protocol-validation evidence unless the entire procedure were wrapped in a separate nested-validation design.

Target-size study policy

`TargetSizeStudyPolicy` is the sole target-size decision authority. It consumes the qualified size population and target-only development/model-selection evidence. Replay semantics and monitor identity remain part of the common frozen training protocol, but replay metric values are diagnostic and are not consumed by target-size ranking, qualification, rejection, or tie-breaking.

Monitor policies are type-distinct from target-size policy. A target monitor of 256 configurations and a replay monitor of 512 configurations, if those values are current, remain monitoring evidence sets; their integers do not create target-size rungs.

Exact continuation fidelity

Each candidate follows one authenticated training continuation:

```
0 -> n1 coarse -> n2 short -> n3 final screen
```

`0 < n1 < n2 < n3` are the exact screening boundaries; `n` is an independent positive production maximum and may equal, exceed, or be less than `n3`. The terminal boundary may be derived internally for the deterministic continuation schedule, but it is not a second screening authority. The short state authenticates the exact coarse model, optimizer, RNG, and protocol parent; the final screen continues the short state. All size candidates use the same foundation, replay semantics, objective, optimizer/LR schedule, exposure policy, precision/backend, and ordered training-seed set. The seed authority is the ordered `seeds` field of the sole enabled training method; current generated campaigns default that field to `[1, 2]`. The target-size policy authenticates this ordered set rather than owning an unrelated seed convention.

Ordinary target-success early stopping is disabled during this experiment because size candidates must be compared at common fidelity boundaries. Each expected candidate/seed resolves to exactly one authenticated stage outcome: strict finite endpoint success or explicit candidate-specific TRAIN2/EVAL2 numerical failure. Generic execution/resource/input/lineage failures remain fail-closed campaign errors. Normal production/CV stopping resumes once the size experiment is complete.

Successive-fidelity funnel

Let $q = |\mathcal{Q}|$. The production decision requires at least three qualified sizes:

```
q < 3      -> insufficient_qualified_sizes
q >= 3     -> coarse n1:       q -> min(q,4)
              short n2:       <=4 -> 2
              final screen n3: 2 -> 1
```

Candidate comparison uses paired seed-aggregated evidence: every candidate uses the same authenticated ordered seed set, avoiding comparisons between unrelated stochastic realizations. Missing, duplicated, reordered, or candidate-specific seed populations fail closed.

At `n1` and `n2`, the coarse practical-equivalence width defaults to 1 meV/Angstrom in the primary target-force metric; the smaller size is preferred inside the configured band. At `n3` the final width independently defaults to 1 meV/Angstrom. Both widths are configurable positive finite policy fields and therefore participate in target-size policy identity; changing either width invalidates reuse of derived study evidence from the old policy. The early screens rank relative promise; they do not require the final absolute force-accuracy threshold.

At `n3`, size selection remains a target-only ranking decision over candidates already admitted by MVQUAL. MVQUAL is the sole hard target-size eligibility authority; the size study does not re-apply target-threshold, replay-retention, energy/stress, physical-integrity, relaxation, deployment, or other downstream model/protocol acceptance gates. Only candidates with complete paired successful endpoints are rankable; authenticated scientific failures remain explicit failure outcomes. Replay scores cannot rank, qualify, reject, or tie-break sizes. Model/protocol acceptance runs only after `N_selected` is frozen and cannot feed back into the size decision.

Typed terminal outcomes

The size study returns a typed decision, not merely an integer:

```
selected(N)
insufficient_qualified_sizes
insufficient_comparable_candidates
nonconverged_at_fixed_ceiling
```

`insufficient_comparable_candidates` records the failed fidelity stage and authenticated candidate/seed failure reasons when authenticated numerical/scientific trajectory failures leave too few complete paired candidates for the required comparison. Input, lineage, and programming errors remain fail-closed exceptions.

If 16,384 reaches the final comparison and remains materially better than every smaller complete finalist by more than the configured final practical-equivalence width, the outcome is `nonconverged_at_fixed_ceiling`. The architecture reports that terminal state rather than synthesizing a larger or intermediate rescue size.

The architecture never creates an intermediate size merely to avoid reporting non-convergence.

Production screening versus algorithm qualification

Ordinary production executes the successive-fidelity funnel and stops training eliminated sizes.

A release/algorithm qualification may retrospectively train the complete candidate population to the configured full horizon `n` to measure survivor recall or calibrate the screening policy. That exhaustive matrix is qualification evidence, not a permanent production requirement. If representative qualification shows that the early screens do not reliably retain eventual finalists, the screening policy must be revised explicitly rather than forcing every campaign to repay every eliminated candidate.

Bounded scientific materialization

The fixed eight-size population is a scientific policy, not a storage mandate. Per training domain, the intended product-scale state is:

```
one fitted selection-input authority
one exact MVIDX authority
one MVSEL2/REPAIR2 master order
prefix metadata for candidate sizes
MVQUAL evidence for required prefixes
training artifacts only for candidates authorized to train
```

After `N_selected` is frozen, each final-development or CV gradient-training DATA7 materialization consumes exactly its local `R_d[:N_selected]` prescribed prefix. DATA7 may fit/materialize training inputs, but it does not rerun an independent membership selector for this protocol.

The architecture specifically rejects eight independent descriptor, sparse-graph, selector-state, or target-dataset copies. Execution caches are reconstructible and bounded.

Scientific non-negotiables

Execution optimization cannot authorize:

- approximate neighborhood semantics in place of the exact frozen relation;
- relaxed hard coverage or obligations;

- changed leakage/correlation boundaries;
- a second target-membership selector;
- independently repaired rungs;
- held-out or locked evidence controlling target size;
- generated/intermediate rescue sizes;
- non-deterministic authoritative rank/repair decisions;
- migration of unsupported historical selector/repair state into current authority.

Any change to these semantics requires an explicit architecture/specification revision rather than an execution optimization.

Part VI - Bounded execution, restart, and performance architecture

Purpose and authority

Execution optimization is acceptable only when it preserves the scientific/statistical authorities defined in Parts I-V and improves measured throughput, memory behavior, storage behavior, or restart cost. Utilization is diagnostic; scientific digests, exact decision traces, and authoritative records decide correctness.

Worker count, queue depth, query-block size, cache location, file-backing threshold, storage path, and similar execution choices do not enter scientific identity unless a current specification explicitly makes them part of the scientific algorithm.

The central rule is:

> change how exact work is scheduled or represented, not what scientific evidence is consumed or what authoritative decision is produced.

Work/span and single-level parallelism

For serial work T_1, critical path T_∞, and P admitted CPU lanes,

$$T_P \geq \max\left(\frac{T_1}{P}, T_\infty\right).$$

Independent work is exposed at the highest useful level. Nested numerical parallelism is suppressed while outer work can fill the resource budget:

$$P_{\text{outer}} P_{\text{native}} \leq P_{\text{budget}}.$$

For native kernels such as cKDTree, BLAS, or OpenMP, a campaign resource scope owns native-thread admission. Individual workers do not independently oversubscribe the machine.

DATA6-to-DATA8 materialization boundary

DATA6 is the last preparation stage that owns the MACE accelerator model. After its final descriptor/prediction consumer, production explicitly releases calculator/model references and unused CUDA allocator state. DATA7/DATA8 are CPU/I/O stages and advertise no GPU jobs. Heavy frame-cache restoration and foundation-energy reconstruction remain lazy until a materialization variant actually misses the completed-artifact reuse path.

DATA7 exposes canonical final/fold domains as the outer parallel unit. Each domain retains independent fitted scaler, PCA, E0, weighting, selection, and coverage state. Immutable frame arrays and authenticated descriptor shards may be shared, but task-local mutable extraction state is not shared between concurrent domains. Outer DATA7 work is admitted through the deterministic resource queue using the live runtime CPU budget and a conservative peak incremental-memory estimate; inner BLAS/OpenMP/PyTorch widths are one while multiple domains are available. Workers publish authenticated immutable DATA7 cache generations and return compact receipts. Only the coordinator mutates production records/checkpoints, in canonical domain order.

Target-size screening is treated as a distinct reuse topology. Campaign preparation owns one nonpersistent namespace resolver that maps each canonical DATA7 `training_domain_digest` to its authenticated coverage authority and materializes REPAIR2 prefixes lazily only for requested `(training domain, size)` pairs. Synthetic coverage IDs are confined to that lookup boundary; DATA7 plans remain keyed by canonical training-domain digest and final-development evaluation cohorts by source label-domain ID. The fixed pre-selection evaluation complement is always the authenticated final-coverage frame set minus the maximum-qualified prefix, so the comparison cohort cannot shrink across `n1/n2/n3` continuation. `ProductionMaterializationPlan` itself rebuilds the canonical DATA5/CV feature-domain topology; callers cannot inject a convenient alternate domain set.

The expensive DATA7 fitted metric/E0/weight core is selection-size invariant, so target-size variants may reuse that core through a reconstructible execution-only index to a fully authenticated DATA7 carrier artifact. The fitted-core index authenticates both the execution recipe and the actual fitted-result digest; publication is create-once/validate-winner, and divergent results for one recipe fail closed. A stale carrier that fails the exact foundation-prediction/reference/lineage reuse contract is discarded and refit rather than promoted to a scientific failure. Reuse admission uses a separate conservative RAM estimate for carrier load, selection/coverage realization, and archive output instead of charging the hypothetical full fit. Size-specific selection and coverage are then realized normally, and the resulting full `Data7PreparationBundle` remains the sole scientific authority. Full shared DATA7 publication likewise requires any concurrently computed winner to match the local bundle digest and deterministic archive SHA. The shared full-artifact recipe is v2 and excludes DATA8-only evaluation membership/target-study outcome state while retaining the exact prescribed training prefix and selection policy. Legacy full-artifact v1 recipes remain read-compatible; reconstructible fitted-core v1 indices are cache misses.

DATA8 separates immutable fixed-file production from production-tree assembly. Unique ExtXYZ cache misses are enumerated first, then balanced across fresh CPU-only interpreter batches when the estimated byte volume is large enough to amortize fresh-interpreter startup; small batches remain on the serial producer. The large read-only context is serialized once with mmap/file references; worker messages carry compact context paths and recipe digests rather than dense arrays. Fixed-file cache generations use atomic publish-or-validate-winner semantics. CPU, RAM, task count, configured free-disk reserve, and both estimated and measured worker-context spill bound concurrency; insufficient transient headroom reduces execution to the serial producer before subprocess launch. After cache population, the production tree, YAML, scripts, protocol identities, tree digest, and promotion are assembled canonically in the parent process.

Externally owned foundation, selected-head, and replay inputs cross an authenticated inode-independent copy boundary into mdstats-owned content-addressed snapshots before reuse. Hardlinking is reserved for mdstats-owned immutable snapshots/cache generations and their consumers. Optimizer-invariant weighted replay and MLCV TRUE_DFT replay-light realizations are content-addressed execution caches, so seed variants do not repeat identical corpus transformations/scans. Shared DATA7/DATA8 caches are reconstructible execution state. Cache layout, worker count, batch assignment, and completion order do not enter scientific identity. Legacy DATA7 flat cache generations and PAR1 lexically ordered checkpoint digests remain read-compatible; current writes use atomically installed content-addressed generations and canonical `plan.domains` checkpoint order.

Deterministic resource-bounded work queue

CPU-heavy independent tasks use a shared deterministic queue abstraction with explicit CPU and memory ownership. Its architectural responsibilities are:

- bound executing, ready, in-flight, and buffered work;
- reserve persistent memory before admitting temporaries;
- propagate deterministic task identities and exceptions;

- allow arbitrary completion order where scientific order is irrelevant;
- restore canonical reduction/commit order where FP64 arithmetic or record order is authoritative;
- expose progress/resource telemetry without placing telemetry in scientific identity.

Task submission may run ahead of execution to hide hand-off latency, but simultaneous execution remains bounded by the declared resource scope.

One product-scale authority per semantic input

The fixed target-size population does not permit one full descriptor/graph/selector copy per rung. Per required training domain, the product-scale execution model is:

```
one fitted DATA7 selection-input authority
one exact neighborhood/MVIDX authority
one MVSEL2/REPAIR2 master order
one compact MVSTATE2 continuation state when persisted
prefix metadata/views for candidate rungs
MVQUAL evidence for required prefixes
training artifacts only for candidates authorized by the size funnel
```

This is an architectural resource invariant, not merely an optimization preference. A realization whose memory/storage scales approximately with eight independent copies of the target-selection state is non-conforming even if it eventually produces the same scientific result.

Exact neighborhood production and reuse

The exact neighborhood engine is the sole geometric producer for the multi-view relation. Query blocks may execute independently through the bounded queue, but completed blocks are committed in the canonical witness/family order required by the scientific representation.

The resulting sparse store is authenticated by semantic inputs such as candidate/reference ordering, family identity, fitted metric/scaling, radius/tolerance semantics, and cache-format identity. Worker count, block size, queue depth, and cache location are excluded.

MVIDX1 consumes this authenticated exact relation and does not repeat geometry when a compatible forward relation exists. Missing, stale, corrupt, or incompatible reconstructible state is rebuilt from the same authoritative inputs.

Deterministic out-of-core MVIDX

Large candidate-to-witness inversions must be bounded in anonymous RAM. Exact row-chunk transpose/inversion and file-backed typed arrays are permitted when they produce the same authoritative sparse arrays as the in-memory realization.

The execution layer therefore preflights RAM and scratch/disk capacity, bounds concurrent inversions, and may use mmap/NPY-compatible arrays for large products. Candidate offsets and candidate-to-witness indices use the current specified canonical types and ordering.

Chunk size, file-backing threshold, and inversion concurrency are execution-only. In-memory and out-of-core paths must satisfy the same scientific content/equivalence contract.

MVSEL2 exact forward/lazy execution

MVSEL2 rank authority remains sequential because each accepted candidate changes the scientific state. Execution acceleration is concentrated in exact candidate-row evaluation, staged lexicographic filtering, vectorized sparse gathers, and a certified lazy representative frontier.

A stale lazy score is only an upper bound. A candidate can be excluded without refresh only when the bound proves it cannot defeat the best exact contender under the frozen tolerance. Otherwise it is refreshed exactly. Authoritative contender comparison and state mutation remain canonical.

MVSEL2 maintains compact witness multiplicity, covered mass, obligation/correlation counts, and representative state. It does not require complete product-scale candidate marginal arrays or normal-path witness-to-candidate inverse propagation.

REPAIR2 exact active-shell execution

REPAIR2 keeps authoritative repair iteration, admissibility predicates, objective/tie hierarchy, accepted/rejected trace, and winner application deterministic.

Independent proposal scoring inside one immutable pre-swap state may execute concurrently when measured work justifies it. Vectorized candidate-row gathers, stamp arrays, bounded-ID indexing, and fused sparse scans are valid preparation optimizations. The accepted winner and authoritative mutation are committed in the frozen deterministic order.

Repair operates on the one master order and protected prefixes. It does not create independently repaired dataset copies for individual target-size rungs.

MVSTATE2 restart boundary

MVSTATE2 is the only current selector/repair continuation-state family. Its persistent bundle is compact and reconstructible from the selected prefix plus authenticated primitive identities.

Persisted state may include:

- selected order/prefix identity;
- per-family witness multiplicity and covered mass;
- obligation and correlation counts;
- representative utility/state required by the current exact continuation contract;
- manifest identities binding dataset/domain, MVIDX, policy, ordering, and schema.

Ephemeral lazy queues and product-scale candidate marginal arrays are reconstructed rather than treated as durable authority.

Restoration authenticates payloads and recomputes continuation invariants from the selected prefix. State from unsupported historical generations is not deserialized or migrated into current authority; the current state is reconstructed from current inputs or the campaign is re-prepared.

MVQUAL independent execution

Same-prefix MVQUAL jobs are independent and may execute concurrently. Their completion order is non-authoritative; persisted comparison order is canonical.

MVQUAL may share authenticated primitive sparse inputs but recomputes the hard predicates independently of selector/repair counters. Parallelism, vectorization, or cache reuse cannot turn selector-internal state into qualification evidence.

Target-size funnel materialization

The configurable `n1/n2/n3` size study materializes training state only for candidates still authorized by the production funnel:

```
qualified population
  -> coarse (`n1`) candidates
  -> at most four short (`n2`) continuations
```

```
-> two final-screen (`n3`) continuations
-> one selected size or typed failure
```

Continuation authenticates model, optimizer, RNG, and protocol parentage. Eliminated candidates are not trained further in ordinary production.

Exhaustive training of the full candidate population to final fidelity is release/algorithm qualification only and must use a bounded dedicated qualification design. It is not permitted to become an unbounded default campaign artifact generator.

Replay indexing and bounded parsing

The selected replay source remains external scientific authority. A replay source index may store authenticated source-byte identity, frame offsets/lengths, atom counts, and source-order geometry identity to permit sparse monitor access and bounded chunk parsing.

The index is reconstructible execution state. It cannot replace replay source, split, label, prediction, monitor, or retention authority. Source mutation or index corruption causes safe reconstruction.

Parser concurrency is introduced only when measurement on representative workload shows benefit and exact persisted replay bytes/identities are preserved.

Training, evaluation, and verification concurrency

Independent training, checkpoint-evaluation, and deployment-verification jobs may execute concurrently under common CPU/RAM/GPU/VRAM admission where their owning policies permit concurrency.

Runtime concurrency never enters the scientific checkpoint score or admissibility policy. Hard GPU/VRAM or RAM limits fail closed rather than silently switching precision/backend, shrinking scientific evidence, or changing model policy.

Positive accelerator qualification is evidence. The architecture does not assume an accelerator path is correct merely because it is available.

Canonical staged checkpoint evaluation and target-size reuse

OPT-EVAL4 owns checkpoint-evaluation execution as a bounded CPU-prepare -> accelerator -> CPU-finalize pipeline. TARGET-SIZE-V5 exact-boundary EVAL2 uses this same scheduler rather than a private checkpoint loop. The target-size parent enumerates and authenticates scientific endpoint authority, the staged workers perform computational preparation/inference/finalization, and the parent validates returned run/checkpoint/target-role/prediction/metric identities before any durable endpoint publication. Cache-only and freshly computed endpoints converge through that same parent validation path, and the target-size reducer cannot run until the complete expected `(size, seed)` population has authenticated terminal evidence in deterministic order.

One compatible target role may expose a stage-resident immutable target context. Reuse is content-addressed for computation but never substitutes byte identity for scientific authority: every contributing artifact lineage is authenticated against the role and exact frame-UID sequence. The stage RAM ledger charges shared target atoms/evaluation views once; per-endpoint admission charges only incremental prepared state. Downstream mutation requires a private copy rather than mutating the shared context.

The accelerator stage retains one resource owner. A TARGET-SIZE-V5 population may serially reuse one worker-private MACE provider/model shell when checkpoint bytes authenticate and exact model class/state key/shape/dtype plus runtime-architecture policy prove weight replacement compatible. Foundation-model providers, CuEq/OEq transforms, compiled providers, structural incompatibility, or other unqualified shells rebuild normally. Corruption or authority mismatch remains fatal rather than falling back. Weight-dependent calculator/descriptor state is invalidated on replacement; geometry graph caches remain separately governed by geometry/policy identity.

Static-inference calibration is execution state, not scientific model identity. A calibrated runtime profile may be reused across checkpoint weights only when the provider exposes the same authenticated weight-independent runtime-architecture digest and the exact authenticated geometry workload, device, dtype, head, acceleration/precision policy, and relevant hardware identity match. Without stable geometry identities, compatibility remains checkpoint-exact. Every use still applies live RAM/VRAM clamping and existing OOM/backoff policy.

Cancellation stops new staged admission and is polled at safe preparation/materialization/inference/finalization boundaries. Owned legacy checkpoint-reconstruction subprocesses are monitored so cancellation or timeout terminates the owned process group and cleans attempt-local staging without publishing partial scientific state. Already authenticated terminal evidence remains restartable.

Memory and storage budget

Long stages account for persistent and transient memory:

$$M_{\text{stage}} = M_{\text{persistent}} + M_{\text{inflight}} + M_{\text{buffered}} + M_{\text{sparse}} + M_{\text{result}} + M_{\text{scratch}}.$$

New work is admitted only when its CPU and memory reservations fit the stage budget. Large reconstructible arrays may use mmap-compatible/file-backed persistence to lower peak RSS or restart cost.

Every persistent cache authenticates its semantic inputs and payload independently. Corrupt/stale reconstructible caches are rebuild events; they are not silently accepted and are not scientific evidence unless another current contract explicitly defines them as such.

Scratch-space admission is part of bounded execution. A stage that can create product-scale temporary files must predict and cap scratch use before production work begins.

GPU/VRAM admission

GPU jobs are admitted against explicit free-memory and configured-budget evidence. Calibration/measurement windows and utilization estimators are runtime policy owned by the current execution specifications, not by release chronology in this manual.

Soft GPU-utilization and fractional-VRAM envelopes regulate additional concurrency above a serial floor. A successfully completed one-job CUDA calibration is direct evidence that serial execution of the applicable job/resource profile is viable, so measured demand above a soft envelope caps additional concurrency at one (serial fallback) instead of proving the queue infeasible; only actual execution failure or genuine device/resource unavailability may terminate queued work. Absence of preflight GPU telemetry selects conservative serial execution without parallel expansion evidence, rather than blocking the first execution attempt when the CUDA device is available.

An execution controller may reduce job concurrency after measured resource pressure. It cannot change the scientific batch/exposure semantics, precision policy, checkpoint evidence, or target/replay membership merely to fit memory unless the owning scientific specification explicitly permits that change.

Adaptive OOM recovery is acceptable only when the recovered execution is protocol-equivalent and the changed execution parameter is non-semantic.

NUMA-ready locality

A flat queue is appropriate when locality is not limiting. Multi-socket systems may add node-local queues/shards, worker affinity, local stealing first, and cross-node stealing to avoid idle lanes.

NUMA policy is an execution extension. It is activated only after measurement and cannot alter scientific identity, canonical reduction order, or data partition/evidence roles.

Vectorization and allocation discipline

Performance-critical implementations should avoid:

- repeated linear searches and immutable-map reconstruction;
- repeated full-array scaling when a fitted/scaled authority can be reused;
- unnecessary concatenation or Python-object materialization where typed arrays suffice;
- repeated per-frame/per-species masks that can be safely cached;
- full candidate rescans when exact sparse/local updates suffice;
- duplicate descriptor/graph/materialization per target-size rung.

Useful exact kernels include offset-derived ragged CSR gathers, bounded integer indexed counting/reduction, epoch/stamp arrays, preallocated typed outputs, and cached static reduction metadata.

Optimization reviews must distinguish arithmetic preparation from authoritative arithmetic order. Reordering memory accesses or batching independent work is acceptable only when the authoritative records satisfy the required exact/tolerance contract.

Progress and observability

Every long-running stage exposes both scientific progress and resource/executor state. At minimum:

1. completed/total work and percent where meaningful;
2. elapsed and ETA when estimable;
3. throughput with an explicit stable unit;
4. active/pending/buffered work or equivalent scheduler state;
5. resource pressure or current hot item when relevant.

A heartbeat is emitted during long periods without task completion. ETA is based on globally committed work.

User-facing MLFF elapsed and known ETA use fixed `HH:MM:SS` formatting; unavailable ETA is `--:--:--`. Presentation state never enters scientific digests or cache identity.

Performance qualification

A performance change is reviewed against representative target-scale work. Evidence records, as applicable:

- wall and CPU time;
- throughput and measured occupancy/utilization;
- peak RSS/VRAM and scratch/persisted bytes;
- queue occupancy/backpressure;
- output/content digests;
- exact scientific-record equality or the explicitly declared tolerance contract.

Sequential-authority algorithms are checked at sufficient state granularity to detect drift—for example MVSEL2 accepted ranks, REPAIR2 swap trace/state, and MVIDX canonical sparse arrays.

Detailed before/after measurements, failed optimization experiments, release qualification results, and chronology belong in benchmarks/audits/history rather than the current architecture.

Part VII - Ownership and extension boundaries

One-generation authority model

The current MLFF workflow has one semantic generation. A record, policy, or artifact is either compatible with that generation or unsupported. Historical selector, repair, migration, and campaign-generation formats do not form alternate current execution paths.

Architecture owns durable structure and scientific/algorithmic invariants. Narrow specifications own exact schemas, policy values, numerical tolerances, failure codes, and module-local runtime behavior. Workplans coordinate proposed transitions and never become product authority merely because an implementation follows them.

The core authority chain is:

```
source evidence and labels
    -> eligibility / conditions / evidence roles
    -> fold- or final-training domain
    -> DATA7 fitted selection inputs
    -> FEAS1
    -> MVIDX1
    -> MVSEL2
    -> REPAIR2 / MVSTATE2
    -> MVQUAL
    -> TargetSizeStudyPolicy
    -> selected protocol-global target size
    -> domain-local selected target prefixes
    -> training / checkpoint selection
    -> held-out protocol validation
    -> final committee / deployment
    -> calibration / activated locked tests / observable validation
```

There is no branch from this graph to MVSEL1, REPAIR1, MVSTATE-REUSE1, MVMIGRATE, ADAPT-MIGRATE, generated-size rescue, or a second DATA7 membership selector.

Scientific decision ownership

Decision or product	Sole current owner	Consumes	Emits	Explicitly does not own
source/label identity	DATA2-family source and label contracts	immutable source material	normalized labeled-record identity	partition, selection, training
conditions/eligibility	DATA3-family contracts	source records	eligible frames and physical conditions	evidence-role assignment
raw features/events	DATA4-family contracts	eligible evidence, profile/provider declarations	partition-independent raw evidence	fitted metrics or target membership
evidence roles and fold domains	DATA5 partition contracts	cohorts, independence evidence, purge rules	development/monitor/CV/calibration/test roles and authorized training domains	target ranking
descriptors/difficulty inputs	DATA6 contracts	authorized domain evidence, frozen foundation model where applicable	raw/blinded descriptor and prediction products	target membership
fitted selection inputs	DATA7 contracts	one authorized fold/final training domain	fitted transforms/metrics, E0 fits, objective/weights, difficulty and condition/provenance inputs	target-membership order or target size
full-pool feasibility	FEAS1	eligible candidates, hard obligations, exact coverage primitives	feasibility/fragility evidence	subset ranking
exact sparse neighborhood relation	MVIDX1	authenticated feature families, scaling, radii, candidate/reference identities	exact sparse relation and forward runtime projection	selector policy
target ordering	MVSEL2	DATA7 inputs, FEAS1/MVIDX1, selector policy	one deterministic progressive order per domain	independent qualification, target size
repaired target ordering	REPAIR2	MVSEL2 order/state, hard obligations, repair policy	one authoritative repaired master order per domain	target-size choice
continuation state	MVSTATE2	authoritative selected prefix and primitive identities	reconstructible compact continuation state	complete candidate marginal arrays, migration
independent hard qualification [illegible]	MVQUAL [illegible]	authenticated primitive sparse inputs and requested prefixes [illegible]	independently recomputed hard coverage/obligation evidence [illegible]	scientific ranking or size choice by itself [illegible]

A narrow specification may refine how its row is realized, but it cannot create a second semantic owner for the decision.

DATA7 boundary: fitted preparation, not membership

DATA7 is the last fitted-preparation authority before multi-view subset construction. For each authorized fold/final training domain it may fit and publish:

- heterogeneous feature transforms and metrics;
- atomic-reference/E0 fits where applicable;
- training objective and configuration/property weight records;
- training-domain difficulty evidence;
- condition, provenance, event, environment, and diversity inputs needed by subset construction;
- immutable identities linking those products to the domain and protocol.

DATA7 does **not** publish an independent quota/FPS membership decision, a second target-membership ladder, or a target-size decision. Representative coverage, diversity/FPS, environment coverage, protected events, difficulty, and condition balance are expressed as inputs, hard obligations, or objective terms of the one MVSEL2 policy.

After target size is frozen, target-size-controlled DATA7 materialization in every final-development and CV gradient-training domain consumes the authenticated prescribed membership `R_d[:N_selected]`. A `TrainingSelectionPlan` used to record that prescribed materialization is a consumer record, not an independent selector: the quota/FPS selector is not invoked and cannot replace the REPAIR2 prefix.

This boundary prevents two selectors from producing incompatible notions of the target set while preserving the useful fitted/statistical information accumulated in DATA7.

One current multi-view chain

FEAS1 and MVIDX1

FEAS1 diagnoses whether the complete authorized candidate pool can satisfy the frozen hard support and obligation predicates. It does not weaken those predicates to make a requested size feasible.

MVIDX1 owns the exact sparse neighborhood relation used by the selector/repair/qualification family. Scientific identity binds candidate/reference order, feature-family/scaling/radius semantics, and exact sparse content. Execution choices such as worker count, chunking, inversion strategy, mmap layout, or queue depth are reconstructible realization details and cannot change scientific identity.

MVSEL2

MVSEL2 is the sole current target-ordering authority. It produces one deterministic progressive order per training domain under the frozen lexicographic scientific policy. The implementation may use exact forward/lazy acceleration, but any acceleration must preserve the same authoritative candidate choices and FP64 scientific decision semantics.

REPAIR2 and MVSTATE2

REPAIR2 is the sole current repair authority. It produces one authoritative repaired order per domain. Candidate rungs are prefix views of that one order; separate rungs are not independently repaired datasets.

MVSTATE2 is authenticated, compact, reconstructible continuation state. It binds the dataset/domain, candidate and family order, MVIDX identity, weights, obligations, correlation units, selector/repair policy, selected prefix, and schema/version identity. It is not a migration envelope for superseded selector state.

MVQUAL
MVQUAL independently verifies hard coverage and obligation predicates for required prefixes. Selector/repair internal counters are not accepted as independent qualification evidence. MVQUAL may reuse authenticated primitive sparse inputs while recomputing the relevant predicates through its own verification path.

Target-size authority

Distinct size concepts
For required training domain d,

$$N_{\text{available},d} = |\mathcal{D}_{\text{eligible},d}|.$$

The nominal scientific target-size population is

$$\mathcal{N}_0 = \{128, 256, 512, 1024, 2048, 4096, 8192, 16384\}.$$

The common materializable population across all required final-development and cross-validation gradient-training domains is

$$\mathcal{N}_M = \left\{N \in \mathcal{N}_0 : N \le \min_d N_{\text{available},d}\right\}.$$

Independent MVQUAL evidence defines

$$\mathcal{Q} = \{N \in \mathcal{N}_M : \text{all frozen hard requirements pass in every required domain}\}.$$

The selected target-training size must satisfy

$$N_{\text{selected}} \in \mathcal{Q} \subseteq \mathcal{N}_0.$$

`N_available`, a monitor cardinality, a replay cardinality, or an implementation batch/budget count can never become `N_selected` through numeric coincidence.

Domain-local membership and protocol-global size
Each fold/final training domain constructs its own leakage-safe repaired order π_d from only evidence authorized for that domain. For a materializable rung,

$$D_{d,N} = \pi_d[: N].$$

The actual frame membership is therefore domain-local. The final `N_selected` is one protocol hyperparameter shared across every required training domain. Cross-validation validates the complete protocol containing that already-frozen size; held-out fold performance does not choose it.

Hard-coverage monotonicity
Because all rungs are prefixes of one repaired order, increasing N only adds candidates. Under a fixed exact hard-coverage predicate, satisfaction cannot regress solely because N grows. Qualified sizes therefore form a contiguous suffix of the materializable ladder.

A pass/fail/pass sequence is an invariant violation in nesting, identity, qualification, or numerical realization. The size funnel must fail closed rather than work around such a pattern.

Target-size study

`TargetSizeStudyPolicy` is the sole scientific target-size owner. It consumes only authorized target-side development/model-selection evidence. Replay semantics/monitor identity may remain bound through the frozen training protocol, but replay metric values are diagnostics and cannot rank, qualify, reject, or tie-break target sizes. Held-out CV evaluation folds and locked tests remain unavailable to the size decision.

Exact fidelity continuation

Each candidate follows one authenticated continuation trajectory:

```
foundation -> coarse n1 -> short n2 -> final screen n3
```

The short screen authenticates the exact coarse model/optimizer/RNG parent; the final screen authenticates the short parent. `0 < n1 < n2 < n3` are the screening boundaries, while fresh production uses an independent positive maximum `n` with no required ordering against `n3`. Candidates use the same foundation, replay semantics, objective, optimizer/LR schedule, exposure policy, precision/backend, and ordered seed set. That seed set comes from the `seeds` field of the sole enabled training method; current generated campaigns default the owning field to `[1, 2]`. The target-size policy serializes the ordered set and does not invent a second seed convention.

At every target-size boundary the endpoint itself is authoritative: `S(N,n1)`, `S(N,n2)`, and `S(N,n3)` are evaluated at matched fidelity. A better earlier checkpoint cannot replace the prescribed endpoint. This is distinct from post-selection production checkpoint selection, where `N_selected` is fixed and the checkpoint epoch may be optimized over the admissible trajectory.

Ordinary target-success early stopping is disabled during the size experiment because candidates must reach comparable fidelity boundaries. Every expected candidate/seed produces exactly one stage outcome: strict successful endpoint evidence or explicit authenticated candidate-specific numerical/scientific failure evidence. Generic execution/resource/input/schema/lineage failures remain campaign errors. Normal production/CV stopping resumes after target size is frozen.

Public orchestration boundary

The TRAIN2 CLI is a projection of existing scientific/execution authorities, not an additional persistent lifecycle authority. Its stable lifecycle is:

```
prepare -> preflight -> select-target-size -> materialize -> preflight -> train -> evaluate ->
verify
```

`select-target-size` is the sole public owner of the restartable `n1/n2/n3` controlled-fidelity loop. It reuses one screening DATA8 matrix and one matrix-bound preflight across all configured boundaries. Once `selected_target_size` is frozen, `materialize` realizes exactly that size across the final-development and configured CV topology; because this changes the active DATA8 matrix, the earlier screening preflight cannot authorize production training. Public `train` and `evaluate` are then restricted to the selected production/CV workload. `status` and `advance` derive these semantic steps from `TargetSizeStudyPlan`, active DATA8 identities, the preflight matrix digest, and existing execution/evaluation receipts; they do not write a second scientific state machine.

Successive-fidelity funnel

Let $q = |\mathcal{Q}|$. Fewer than three qualified sizes is a typed failure. Otherwise the production funnel is:

```
q >= 3
  coarse n1:       q -> min(q,4)
  short n2:       <=4 -> 2
  final screen n3: 2 -> 1
```

All candidates use the same authenticated ordered training-seed set, and comparisons aggregate paired seed evidence rather than unrelated stochastic realizations. Missing, duplicated, reordered, or candidate-specific seed populations invalidate the state.

At `n1` and `n2`, the configurable coarse practical-equivalence width defaults to 1 meV/Angstrom in the primary target-force metric, and candidates inside that band prefer the smaller size. The independently configurable final width also defaults to 1 meV/Angstrom for `n3` ranking and fixed-ceiling material superiority. Both positive finite values are policy-identity fields, not frozen schema constants. Early screens need not satisfy the final absolute force-accuracy threshold.

At `n3`, the two finalists are ranked only by the target-size study metric under the frozen practical-equivalence rule. MVQUAL remains the sole hard target-size eligibility gate; target-threshold, replay, physical-integrity, relaxation, deployment, and other model/protocol acceptance evidence is downstream of the immutable size choice. Only complete paired successful candidates are rankable; authenticated trajectory failures remain explicit evidence and replay scores cannot affect ranking or tie-breaking.

Typed terminal outcomes

The target-size decision is a typed result, at minimum:

```
selected(N)
insufficient_qualified_sizes
insufficient_comparable_candidates
nonconverged_at_fixed_ceiling
```

`insufficient_comparable_candidates` records the failed fidelity stage and authenticated candidate/seed failure reasons when authenticated numerical/scientific trajectory failure leaves too few complete paired candidates for a required comparison. Input/lineage/programming defects remain fail-closed exceptions.

If 16,384 reaches the final comparison and remains materially better than every smaller complete finalist by more than the configured final practical-equivalence width, the result is `nonconverged_at_fixed_ceiling`. No generated/intermediate rescue size is synthesized.

Exhaustively training all sizes to the final fidelity to measure survivor recall is release/algorithm qualification, not the ordinary scientific production path.

Bounded materialization and execution ownership

The fixed scientific ladder does not imply eight independent product-scale copies. Per training domain, the intended materialization model is:

```
one fitted selection-input authority
one exact neighborhood authority
one MVIDX authority
one MVSEL2/REPAIR2 master order
prefix metadata for candidate rungs
MVQUAL evidence for required prefixes
training artifacts only for candidates authorized to train
```

Descriptor, sparse-graph, selector, and repair caches are reconstructible and resource-bounded. Out-of-core inversion, memory mapping, chunking, work queues, NUMA placement, and concurrency are execution strategies. They may change work/span, RSS, VRAM, scratch, or wall time; they may not change scientific membership, hard coverage, rank order, paired-seed comparison, or decision authority.

This bounded representation is an architectural requirement because duplicating product-scale state per rung makes the fixed ladder scale roughly with rung count and is not an acceptable realization of the scientific policy.

Superseded campaign generations

Current architecture does not provide MVMIGRATE/ADAPT-MIGRATE state machines, legacy construction modes, or alternate v1 selector/repair execution. Artifacts from unsupported generations fail clearly and require campaign re-preparation under the current architecture.

Historical schemas may remain under `docs/history/mlff/` when needed to interpret durable evidence. Their presence never creates current product compatibility requirements.

Physical-observable validation ownership boundary

Physical-observable calculation is not owned by `mdstats.training_data`. RDF, coordination, neighbor-angle statistics, connectivity, topology statistics, MSD, VACF, spectra, VDOS, diffusion, displacement distributions, current correlations, ionic conductivity, and related physical observables remain authoritative in their respective `mdstats.analysis` modules, specifications, and architecture manuals.

The MLFF layer owns only:

1. choosing an advisory observable-recommendation profile and explicit recipe;
2. constructing an immutable recipe of analysis call IDs and parameters;
3. running the same recipe on matched reference and MLFF collections;
4. preserving verified collection/frame-selection identity, symmetric reference/candidate trajectory-generation identity, runtime/capability identity, warnings, and analysis-owned result identities;
5. binding execution to an explicit statistical role and, where required, a predeclared comparison policy, protocol freeze, and test-activation record;
6. applying comparison and acceptance policies only after those policies are frozen and independently identified.

The MLFF layer does not own the physical numerical algorithms, normalization, neighbor definitions, plateau estimators, spectral transforms, or graph statistics.

Compact structural descriptors used for partitioning or subset construction are workflow inputs. Full physical observables used to judge a trained model remain analysis products. Expensive trajectory observables such as diffusion, VDOS, conductivity, or residence statistics are validation jobs rather than ordinary frame-selection features.

Physical-observable evidence has one explicit role such as `training_diagnostic`, `checkpoint_monitor`, `outer_validation`, `calibration`, `locked_test`, or `external_benchmark`. The role is never inferred from filenames. Realized observables cannot choose their own acceptance policy, and locked-test evidence cannot alter fitting, subset construction, target-size choice, training protocol, checkpoint selection, calibration policy, or acquisition.

Extension boundaries

A future extension is compatible with this architecture only when it preserves the ownership graph above or explicitly revises it. In particular:

- a new feature/provider may enrich DATA4/DATA6/DATA7 inputs but cannot create a second membership selector;
- a new subset objective may extend MVSEL2 policy only through an explicit scientific design revision that preserves deterministic single-owner ranking;
- a new size-screen metric may enter `TargetSizeStudyPolicy` only with an explicit evidence role and leakage analysis;
- a new acceleration may change execution representation only after exactness/resource qualification against the same scientific semantics;
- a new campaign generation is not entitled to automatic migration support;

- a custom atomwise/auxiliary loss defines a different `TrainingProtocolIdentity` and requires its own qualification rather than silently modifying the current protocol.

Decision summary

The current MLFF subsystem follows these durable rules:

1. independent evidence remains independent;
2. the complete training protocol is the comparison unit;
3. fitted preparation and target membership are separate authorities;
4. MVSEL2/REPAIR2/MVSTATE2/MVQUAL are the only current multi-view chain;
5. target size and monitor cardinalities are typed, distinct policy families;
6. frame membership is domain-local while selected target size is protocol-global;
7. one repaired master order defines every candidate prefix and hard coverage is monotone with increasing prefix size;
8. the target-size experiment uses development/model-selection evidence, configured `n1/n2/n3` continuation on a screen-local horizon `n3`, paired seeds, and typed non-convergence/failure outcomes; fresh selected-size production owns the independent horizon `n`;
9. MVQUAL is the sole hard target-size eligibility authority; downstream model/protocol acceptance cannot alter the immutable size choice;
10. locked tests remain sealed until the frozen protocol/committee activation boundary;
11. unsupported old campaigns are re-prepared rather than migrated;
12. execution is resource-bounded and may not alter scientific semantics.

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